

CENG381 Stochastic Processes

Irreducibility and Periodicity — Practice 1

Q1 (30 points). Consider the Markov chain on states $\{1, 2, 3, 4\}$ whose transition matrix is the deterministic cyclic shift:

	1	2	3	4
1	[0	1	0	0]
2	[0	0	1	0]
3	[0	0	0	1]
4	[1	0	0	0]

- Draw the transition diagram. Is the chain irreducible? Justify by verifying that every state is reachable from every other state.
 - Find the period of state 1. Recall: the period is $\gcd\{t \geq 1 : P^t(1,1) > 0\}$. List the set $T(1)$ and compute its gcd.
 - Compute P^4 and P^5 . Does the chain converge to its stationary distribution $\pi = (1/4, 1/4, 1/4, 1/4)$? Explain why the period prevents convergence.
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Q2 (35 points). Consider the Markov chain on states $\{0, 1, 2, 3\}$ with:

$$\begin{aligned} P(0,1) &= 1, & P(1,2) &= 1, & P(2,3) &= 1 \\ P(3,0) &= 1/2, & & & P(3,2) &= 1/2 \\ \text{All other entries} & \text{are } 0. \end{aligned}$$

- Write the 4×4 transition matrix P explicitly. Is the chain irreducible?
 - Find the period of state 0. Identify all return times $T(0) = \{t : P^t(0,0) > 0\}$ and compute $\gcd(T(0))$.
 - Find the stationary distribution π by solving $\pi \cdot P = \pi$. What is the expected return time to state 0?
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Q3 (35 points). Take the chain from Q2. Because it has period > 1 , it does not converge. Define the lazy version $Q = (1/2) \cdot I + (1/2) \cdot P$.

- Write the 4×4 matrix Q explicitly.
- Show that state 0 has period 1 in Q , i.e., that $Q(0,0) > 0$ so $T(0)$ contains $t=1$, making $\gcd(T(0))=1$.

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- c) The stationary distribution of Q is the same as P . Verify this: show that if π satisfies $\pi \cdot P = \pi$, then π also satisfies $\pi \cdot Q = \pi$. State the long-run fraction of time the chain spends in each state.